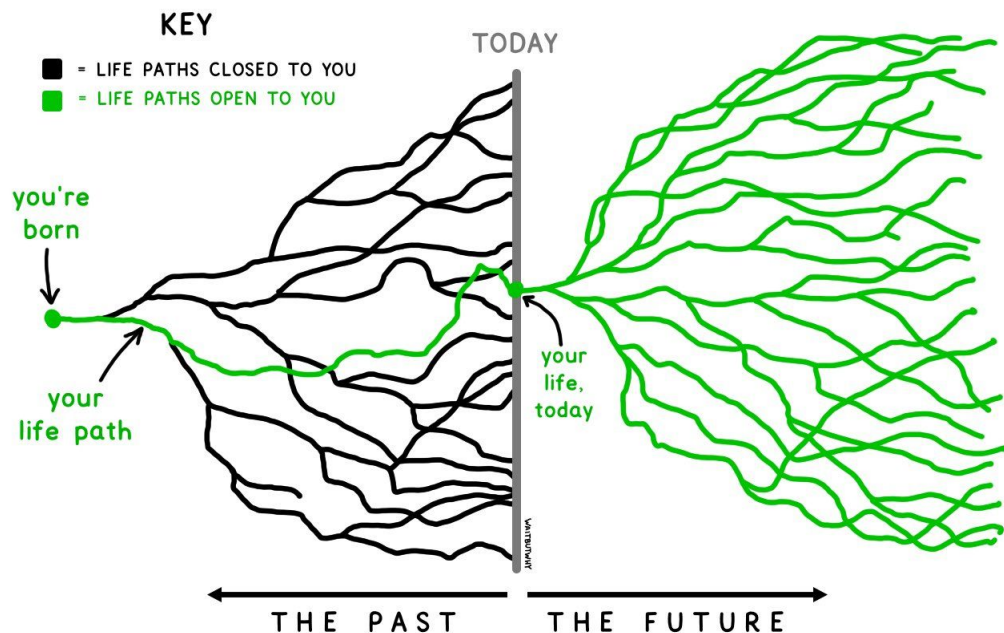




No Matter Who You Are, If You
Done Being A Student, You'll
Quickly Become Irrelevant In
Pages of Time.

So If You Want To Spend Time
“Learning DL” You Better Learn
“Proper DL” (including ML, Statistics,
Optimization, Coding & Stuff...)

Foundations Are Critical!
Rest You Can Learn Along The Way

Deep Learning

CSE641/ECE555

Lecture 1 | Take your own notes during lectures

Vinayak Abrol <abrol@iiitd.ac.in>



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY DELHI

Just gotten over ML



**ML comes back as
DL twice as bad**



Course Overview

I. Objectives

- A. Goals of this course are:
 - 1. understand the foundations of DL
 - 2. get an overview of the techniques used to build DL models
 - 3. get an overview of some subfields of DL

II. Broad Topics

- A. What is Deep Learning & a Deep Neural Network (DNN)?
- B. Supervised Learning & Shallow DNN
- C. Models, Loss Functions & Training Algorithms
- D. Why does deep learning work?
- E. Lots of Maths

Prerequisite

- A. Basic computer science principles
- B. Big-O notation
- C. Comfortably write non-trivial code in Python/C++
 - 1. Start learning TF/Pytorch if you haven't already!
- D. Probability & Random Variables
- E. Expectations & Distributions
- F. Linear Algebra & Multivariate/Matrix Calculus
 - 1. Matrix algebra
 - 2. Gradients and Hessians
 - 3. Eigenvalue/vector

Class time, Evaluation and Books

Class time: Tuesday & Thursday – 4:30 pm to 6 pm

Quiz (N-1): 15 marks

Assignments: 30 marks
4 Programming assignments and 2 Kaggle challenges

Project*: 60 marks

End-sem Exam: 30 marks

Mid-sem Exam: 20 marks

-1 for asking extension
-1 for posting +1 on GC

In Class Quiz: Bonus marks

Attendance: 5 marks

Exception - Medical Emergency (approval required)

Final grading will take into account your consistency in the course.

Last Year's Average Grade: B-

[F Grade on scoring <35% marks + <30% marks in Mid & Endsem]

Books

- Deep Learning by Goodfellow
- Deep Learning by Christopher M. Bishop

Videos lectures

- MIT, Andrew Ng

Classroom Page and Office Hours

★ Course website

- Google Classroom: Deep Learning (2025)

★ Teaching Assistant

- Puneet Singh <puneetb@iiitd.ac.in>

★ My office

- B-409, Research and Development Block
- Office Hours
 - for any doubts and clarifications - Wednesday 10-11am
 - Send me an email first in case you have multiple doubts

★ Academic Dishonesty

- **Institute Policy for Plagiarism and Cheating**

Class Code

aar27fv

THE NOBEL PRIZE IN PHYSICS 2024

Illustrations: Niklas Elmehed



John J. Hopfield Geoffrey E. Hinton

"for foundational discoveries and inventions
that enable machine learning
with artificial neural networks"

THE ROYAL SWEDISH ACADEMY OF SCIENCES

THE NOBEL PRIZE IN CHEMISTRY 2024

Illustrations: Niklas Elmehed



**David
Baker**

"for computational
protein design"

**Demis
Hassabis**

"for protein structure prediction"

**John M.
Jumper**

THE ROYAL SWEDISH ACADEMY OF SCIENCES

What A Time To Be Alive !

Announcing the NeurIPS 2024 Test of Time Paper Awards

Generative Adversarial Nets (Goodfellow et al.) - 85,000+ citations!

This basically kickstarted the whole field of modern generative AI. If you're working with GANs or any kind of generative modeling today, you're building on their foundation.

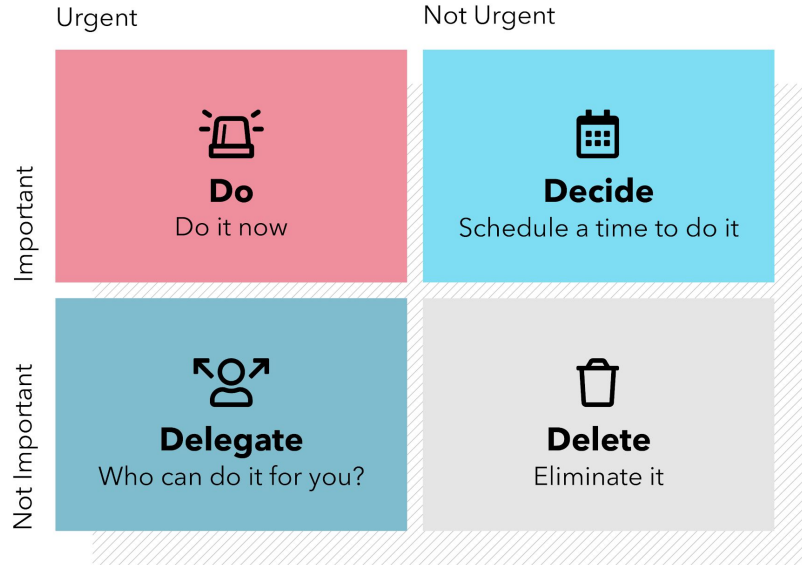
Sequence to Sequence Learning with Neural Networks (Sutskever et al.) - 27,000+ citations!

The encoder-decoder architecture they introduced laid the groundwork for modern LLMs and foundation models.

Crazy to think how far we've come from these papers in 10 years!

DL or NLP or CV or SML ?

The Eisenhower decision matrix



Watching a
Model Train

Watching a
model Train



This is the decade of Generative AI

Majority of the domains have a DL-based state-of-the-art solution

Why/What is Deep Learning ?

By observing, humans can

- Recognize patterns
- Learn from examples
- Solve problems

We learn through

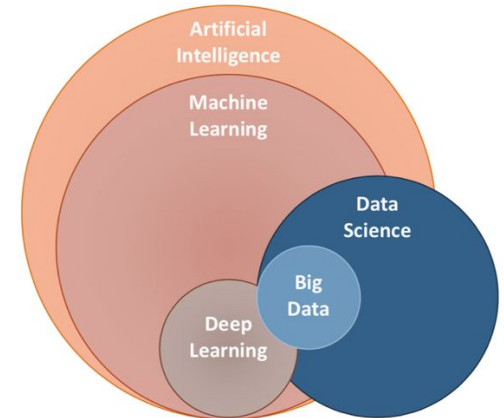
- Association via memorizing important & invariant information
- Building hierarchical relationships

The information is in the connections - **Alexander Bain (1873)**

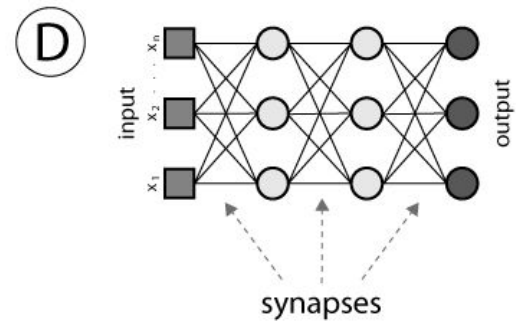
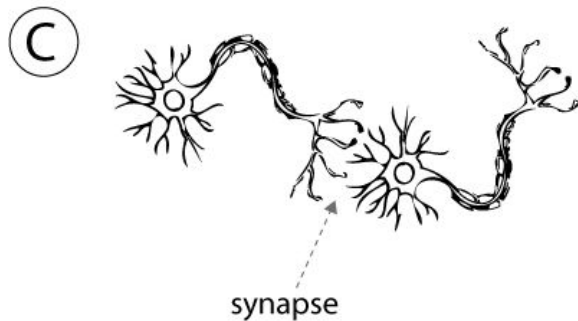
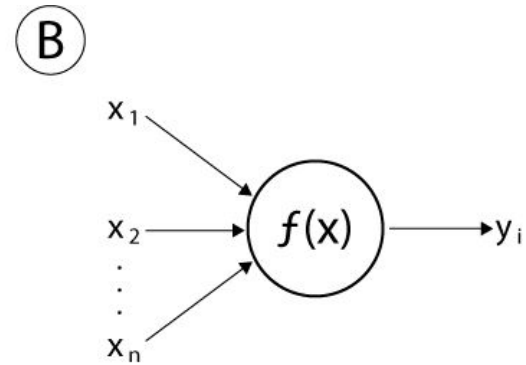
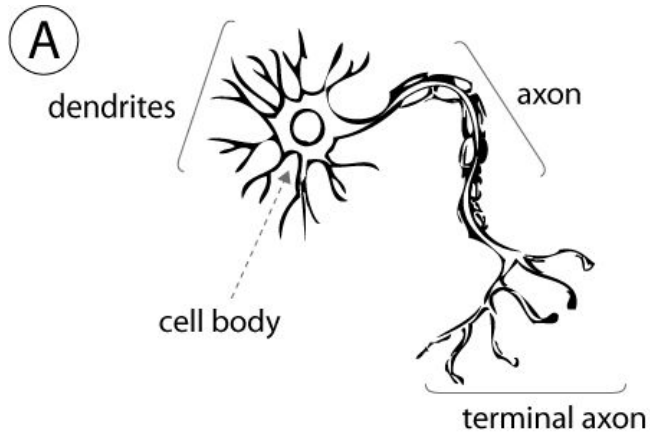
- Different neurons fire depending upon the intensity they receive.
- Connection becomes stronger as they encounter more associations.

AI requires cognitive reasoning.

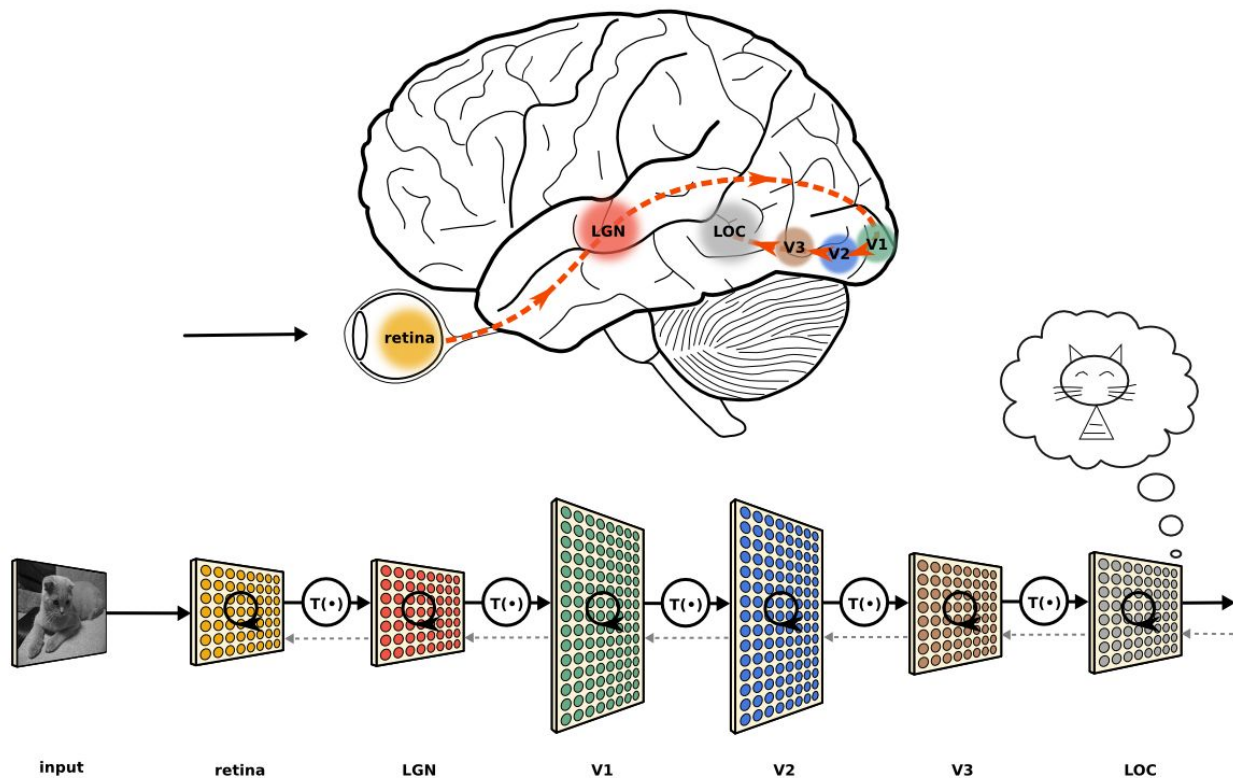
It includes Search, Knowledge, Reasoning, Learning.



Why/What is Deep Learning ?

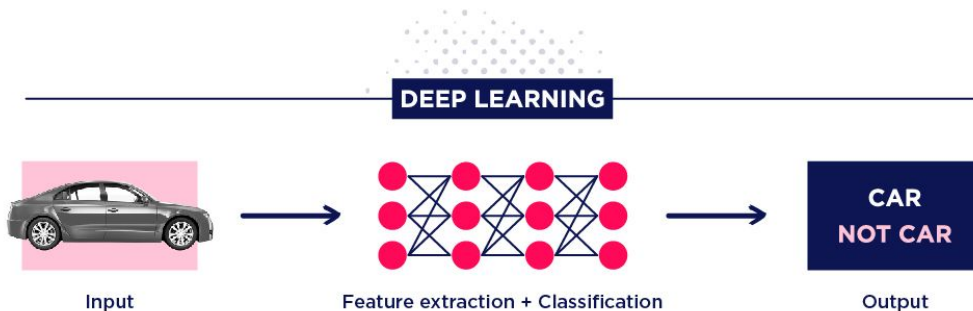
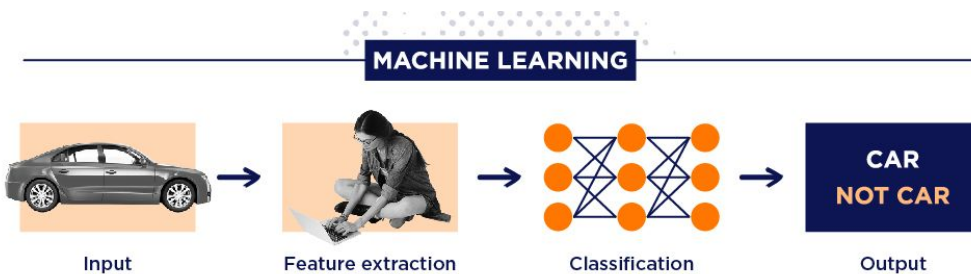


Why/What is Deep Learning ?



Why/What is Deep Learning ?

Caution: Deep Learning is more about Depth [function composition]
And less about end-to-end learning



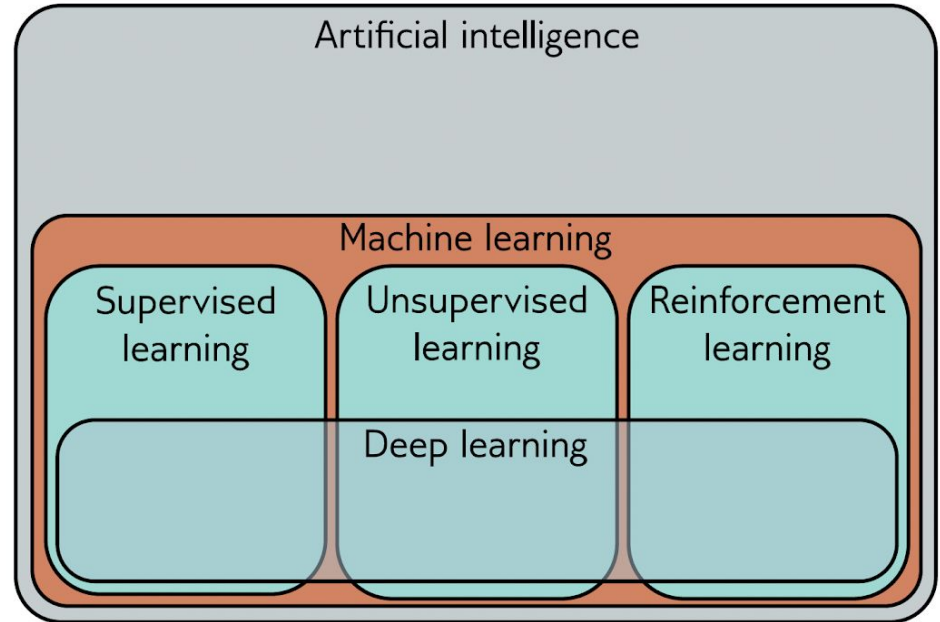
Types of Learning

Knowledge-Based Learning

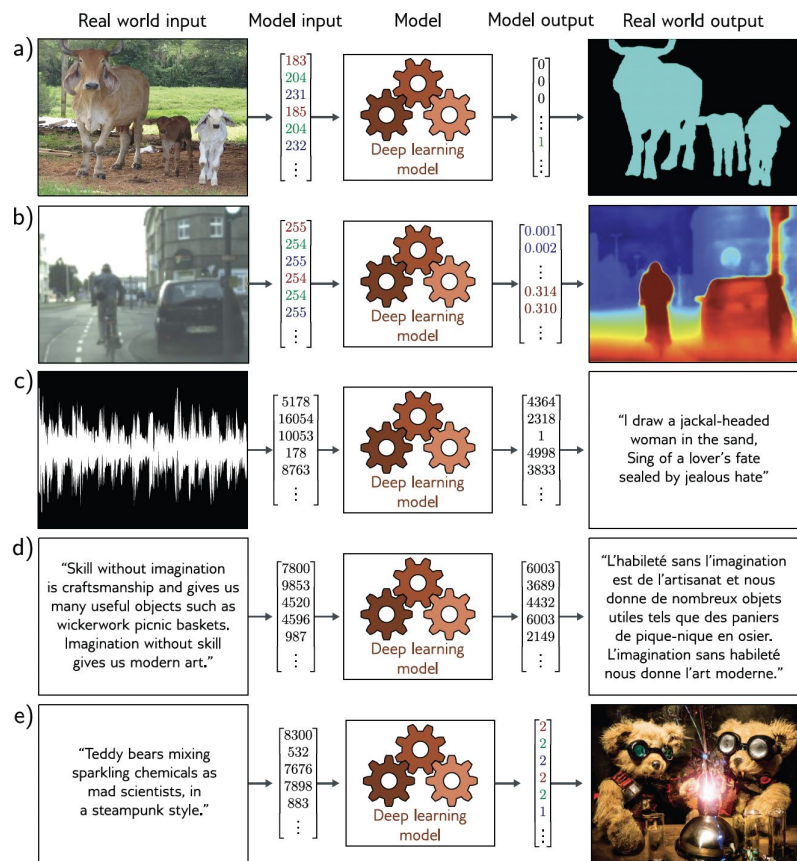
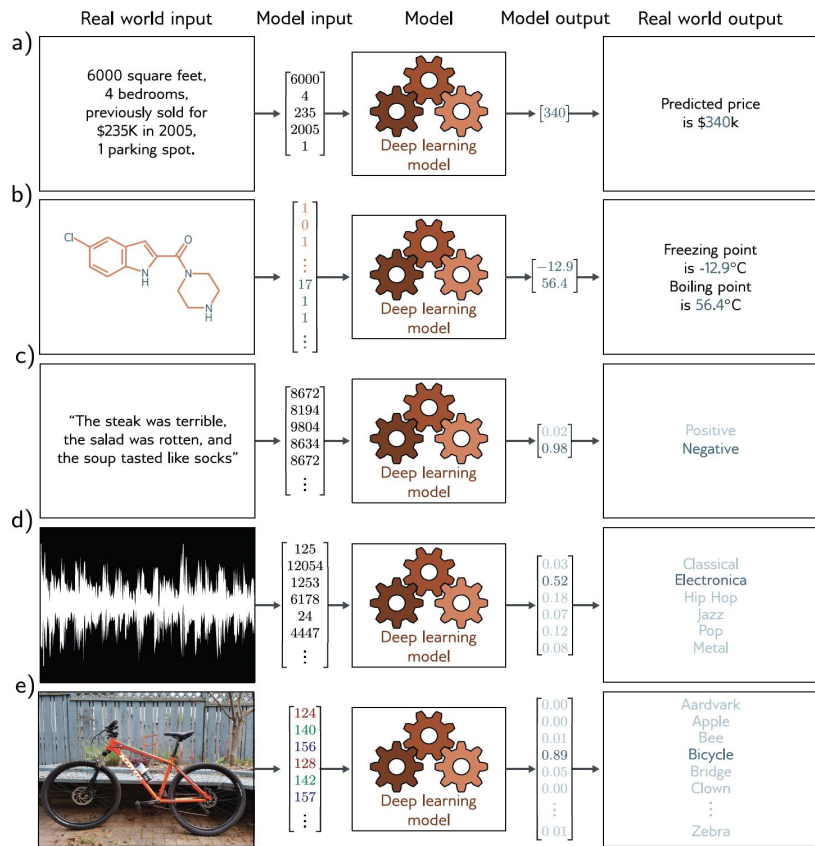
- Inductive vs Deductive

Feedback-Based Learning

- Unsupervised
- Supervised
- Semi-supervised
- Reinforcement



Example Learning Problems & Applications



Example Learning Problems & Applications



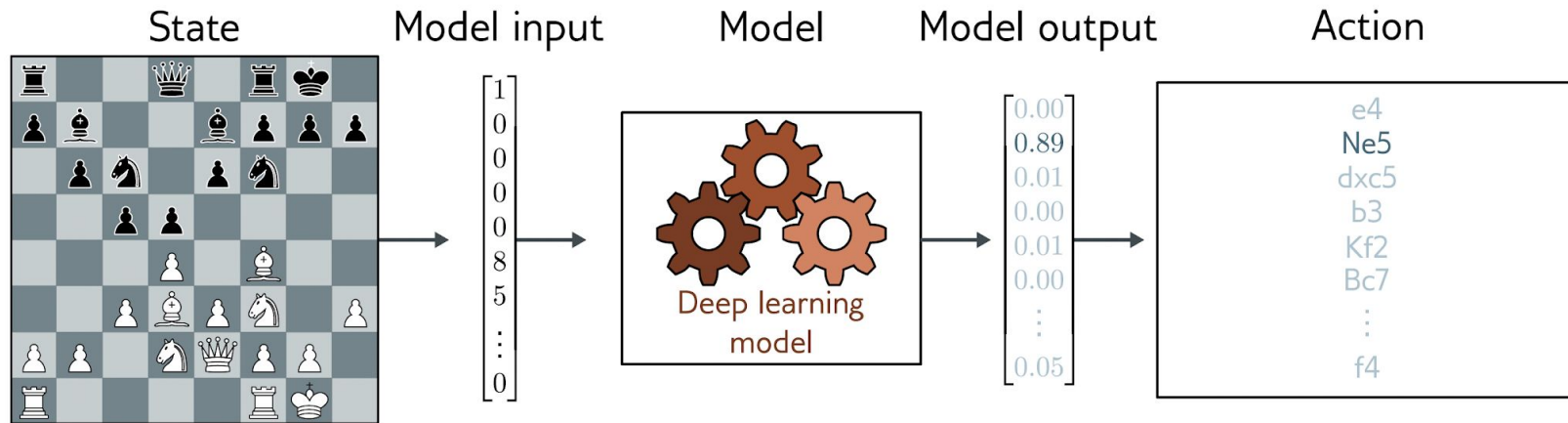
Generative models for images. Left: two images were generated from a model trained on pictures of cats. These are not real cats, but samples from a probability model. Right: two images generated from a model trained on images of buildings. Adapted from Karras et al. (2020b).

Example Learning Problems & Applications



Image interpolation. In each row the left and right images are real and the three images in between represent a sequence of interpolations created by a generative model. The generative models that underpin these interpolations have learned that all images can be created by a set of underlying latent variables. By finding these variables for the two real images, interpolating their values, and then using these intermediate variables to create new images, we can generate intermediate results that are both visually plausible and mix the characteristics of the two original images. Top row adapted from Sauer et al. (2022). Bottom row adapted from Ramesh et al. (2022).

Example Learning Problems & Applications



Policy networks for reinforcement learning. One way to incorporate deep neural networks into reinforcement learning is to use them to define a mapping from the state (here position on chessboard) to the actions (possible moves). This mapping is known as a *policy*. Adapted from Pablok (2017).

Example Learning Problems & Applications



Multiple images generated from the caption “A teddy bear on a skateboard in Times Square.” Generated by DALL·E-2 (Ramesh et al., 2022).

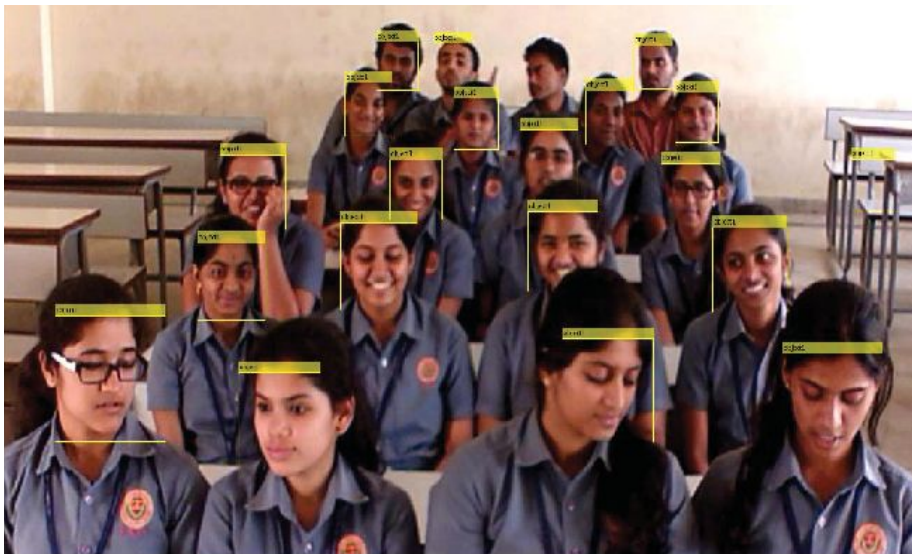
Project* 60%

Automated Attendance Management System (AMS)

- Project can be opted in groups of two.
- Get 60% marks and a minimum of B Grade by building this system using DL. (Only fully functional system)
- Rest 40% marks will be accounted from regular course components after scaling.
- Your solution should ideally scale to a minimum of 500 students per course.

Problem setting:

- You will be provided with multiple smartphone camera images of students attending DL lectures.
- AMS should be end-to-end right from detection, segmentation, recognition, enrolment (database) and verification.
- Realtime fine tuning & testing should be demonstrated on images from a different course.
- You are free to use any tool, library, models, software stack but DL compute requirements are restricted to Colab/Kaggle GPU.



More details will be posted on GC in week 2.

Thanks | Let's Enjoy the Deep Ride Together

No, You Don't Get an A for Effort

An **A** required great work.

Yet today, many students expect to be rewarded for the quantity of their effort rather than the quality of their knowledge.

**Achieving Success is Not
Difficult but Sustaining it is!**

